



AURA — Autonomous AI Wedding Post-Production · Master Build Plan (30 Phases)



Shoot the wedding. Import the RAWs. Click once. Deliver.

AURA is an autonomous AI post-production system for wedding photographers: 1,000–4,000 RAW files in → a culled, edited, retouched, quality-checked, consistently graded, export-ready gallery out. Every decision explained, confidence-scored and reversible.

The deliverable

A complete build bundle written so **Claude Code can implement it without guessing**: 46 files, ~110,000 words, 30 phase specifications averaging **6 pages each** (you asked for 3 — every phase is at least 5).

[AURA-Wedding-AI-Master-Plan.zip](#)

[AURA-Wedding-AI-Master-Plan.zip](#) — 30 phase specs + 15 strategy documents

File	What it gives Claude Code
<code>CLAUDE.md</code>	Operating manual: 9 invariants, phase ritual, repo layout, hard coding rules, cloud-key rules
<code>AGENT-TEAM.md</code>	Your 23-role virtual IT company and the two-hat review rule
<code>00-PRODUCT-STRATEGY.md</code>	Positioning, competitor scorecard, V1 success metrics

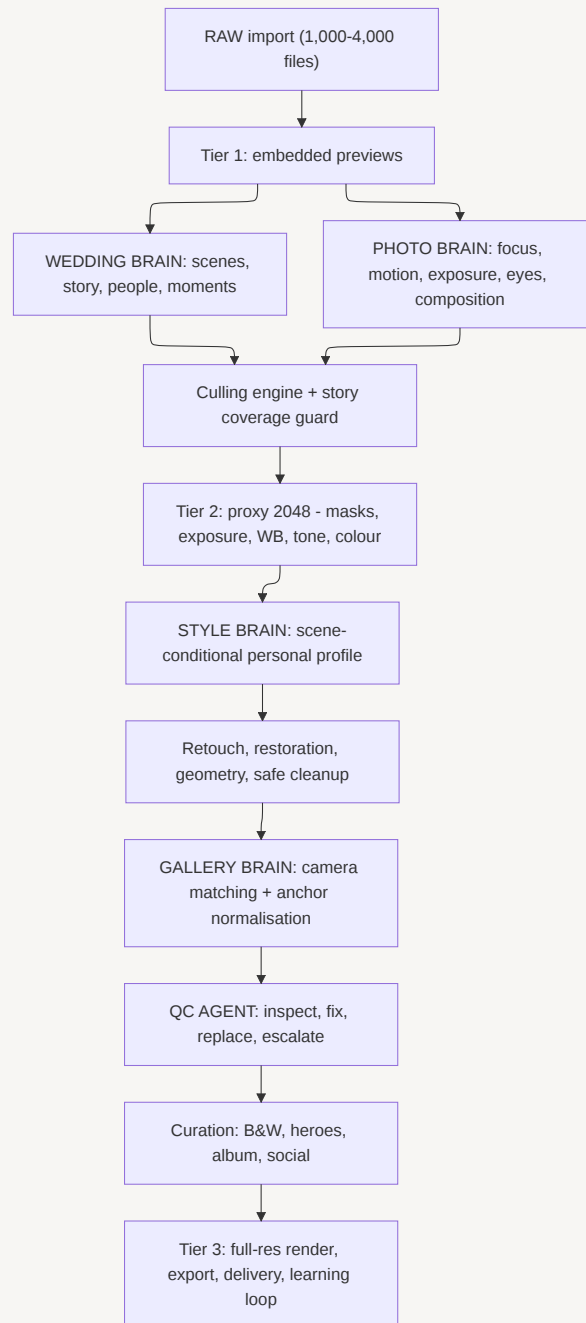
File	What it gives Claude Code
01-ARCHITECTURE.md	Process model, crate layout, three-tier compute pipeline, hardware strategy
02-AI-MODEL-STACK.md	17 local models with sizes and quality gates + the 6 governed cloud tasks
03-DATA-MODEL.md	Full SQLite schema, cache layout, edit-recipe JSON v1, decision ledger
04-ROADMAP-30-PHASES.md	Phase table, epics, version milestones
05-QA-STRATEGY.md	Test pyramid, three fixture weddings, ~20 CI quality gates, chaos tests
06-CODING-STANDARDS.md	Rust / TypeScript / Python standards and the review checklist
07-DATASET-AND-TRAINING.md	The Wedding Intelligence Dataset — the real moat
08-BUSINESS-GTM.md	Pricing, unit economics (89 % gross margin), launch sequence
09-RISK-REGISTER.md	20 ranked risks with owners and triggers
10-GLOSSARY.md	Shared vocabulary to keep 30 phases coherent
phases/PHASE-01..30-*.md	The build itself — one powerful feature per phase
WEDDING-AI-MASTER-PLAN.md	Everything in one file

The stack (decided, not suggested)

Layer	Choice	Why this one
Desktop shell	Tauri 2 + React 18 + TypeScript	Native speed, tiny binary, Rust core in-process
Core engine	Rust (2021)	Memory safety at C speed; ideal for 4,000-file concurrency
RAW decode	LibRaw via FFI	CR2/CR3, NEF, ARW, RAF, DNG, ORF
GPU render	wgpu + WGSL	One shader codebase for Windows and macOS

Layer	Choice	Why this one
Inference	ONNX Runtime → TensorRT / CUDA / DirectML / CoreML / CPU	One model artefact, every machine your customers own
Catalog	SQLite (WAL) + JSON/XMP sidecars	Zero admin, transactional, interoperable
Training	Python 3.11 + PyTorch 2 → ONNX	Standard research stack with parity-verified export
Cloud AI	Your API key , through one governed gateway	Your key, your budget, hard caps, offline fallback

Architecture



The three-tier compute model is the reason this hits **3,000-image analysis and cull in under 8 minutes**: never spend five seconds retouching a frame the culler will reject.

The 30 phases

#	Phase	Epic	Owners	Est.
01	Project Foundation, Catalog & Wedding Project Ingest	E1	CTO, TLC, SRC, SFE, DEVOPS	2 w
02	RAW Decode Engine & Three-Tier Preview Pyramid	E1	TLC, SRC, COL, PERF	2.5 w
03	Inference Runtime & Signed Model Package Manager	E1	MLL, MLOPS, SRG, SEC	2 w
04	Cloud AI Gateway & Agentic Reasoning Runtime (your key)	E1	AGT, CTO, SEC, MBE	2 w
05	Perceptual Embeddings & Wedding Similarity Index	E2	MLL, SRML, SRC	1.5 w
06	Face Detection, Recognition & People Intelligence	E2	MLL, SRML, SRC, SEC	2.5 w
07	Wedding Scene AI & Story Timeline Segmentation	E2	MLL, SRML, AGT, SRC	2.5 w
08	Smart Burst Grouping & Duplicate Detection	E2	MLL, SRC, MLR	1.5 w
09	Frame Integrity AI: focus, motion, exposure, noise, eyes	E2	MLL, SRML, COL, SRC	2.5 w

#	Phase	Epic	Owners	Est.
10	Expression, Emotion & Moment Ranking AI	E2	MLL, SRML, AGT, MLR	2.5 w
11	Composition & Aesthetic AI	E2	MLL, SRML, MLR	2 w
12	Autonomous Culling Engine, Coverage Guard & Gallery Sizing	E2	MLL, SRC, PM, MLR	3 w
13	Explain My Edit, Confidence Calibration & Decision Ledger	E2	MLL, AGT, SFE, SRC	2 w
14	Non-Destructive Edit Recipe & GPU Develop Engine	E3	COL, SRG, TLC, PERF	3 w
15	Exposure AI & White Balance AI (mixed-lighting mastery)	E3	COL, MLL, SRML	2.5 w
16	Tone AI, Adaptive Curves, HSL AI & Skin-Tone Protection	E3	COL, MLL, SRML	2 w
17	Style Learning: scene-conditional "Teach My AI" profiles	E3	MLL, SRML, MLOPS, COL	3 w
18	Local Mask AI: automatic semantic masking	E3	MLL, SRML, SRG	2.5 w
19	Local Light Sculpting: face light, subject,	E3	COL, MLL, SRG	2 w

#	Phase	Epic	Owners	Est.
	background, dodge & burn			
20	Portrait Retouch AI with Natural Texture Protection	E4	MLL, SRML, COL, SRG	3 w
21	Micro-Retouch Suite: hair, teeth, eyes, clothing, glare	E4	MLL, SRML, SRG, COL	2.5 w
22	Restoration Stack: denoise, selective sharpen, face recovery	E4	COL, MLL, SRML, PERF	3 w
23	Geometry Suite: lens corrections, straightening, smart crop	E4	COL, SRC, MLL	1.5 w
24	Generative Cleanup & Distraction Removal (safe by construction)	E4	MLL, SRML, SRG, SEC	3 w
25	Gallery Intelligence Engine: cross- photo colour, skin, scene consistency	E5	COL, MLL, SRC, TLC	3 w
26	Multi-Camera & Second-Shooter Matching	E5	COL, MLL, SRC	2 w
27	AI Quality- Control Agent & Automatic Re-Edit Loop	E5	AGT, MLL, QAL, SRC	3 w

#	Phase	Epic	Owners	Est.
28	Zero-Touch Wedding Autopilot Orchestrator	E5	TLC, EM, PERF, SRC	3 w
29	Curation: B&W, hero photos, album story, social picks	E6	MLL, AGT, PM, SRC	2.5 w
30	Delivery, Integrations, Learning Loop & Release Engineering	E6	DEVOPS, TLC, MLOPS, PM, SEC	4 w

Ship V1 at phases 01–17 + 28 + 30. A shipped culling-and-grading autopilot beats an unfinished everything-machine.

What every phase contains (16 sections)

Mission · dependencies and unlocks · KPIs · competitor benchmark · why it matters · in/out of scope · ASCII architecture diagram · every file to create with its purpose · **frozen Rust/TypeScript/SQL contracts to copy verbatim** · algorithms step by step · cloud-AI task specs with prompt and JSON schema · implementation steps · **a task table assigning every job to a named role with deliverable and estimate** · handoff diagram · test plan · performance budgets · telemetry events · risks with mitigations · acceptance criteria · and a ready-to-paste Claude Code prompt.

Your AI IT company (23 roles)

Code	Role	Owns
CTO	Chief Architect	Architecture, ADRs, invariants, final phase sign-off
PM	Product Manager	Feature definition, competitor parity, user-visible acceptance

Code	Role	Owns
EM	Engineering Manager	Task breakdown, sequencing, blockers, phase exit report
TLC	Tech Lead — Imaging Core	Crate boundaries, public APIs, error taxonomy, pipeline correctness
SRC	Senior Engineer — Core	Catalog, job graph, RAW/IO, concurrency, FFI safety, resume
SRG	Senior Engineer — GPU	Develop engine, shaders, tiling, GPU memory, throughput
MLL	ML Lead — Vision	Model portfolio, training strategy, eval gates, model cards
SRML	Senior ML Engineer	Model implementation, training, quantisation, ONNX parity
MLR	ML Research Engineer	Literature, ablations, loss design, label schemas
MLOPS	MLOps / Packaging	Registry, signing, delta updates, EP benchmarking, model CI
AGT	AI Agent & Prompt Engineer	Cloud LLM/VLM orchestration, tool schemas, JSON validation, cost governor
COL	Colour Scientist	Colour spaces, camera profiles, skin-tone science, dE2000
SFE	Senior Frontend Engineer	App shell, virtualised grid, state machines, typed IPC
MFE	Mid-Level Frontend Engineer	Panels, settings, review queues, i18n, component tests
MBE	Mid-Level Backend Engineer	Cloud services, licensing, uploads, delivery integrations
DATA	Data Engineer / Curator	Dataset ingest, labelling, splits, versioning, leakage prevention
QAL	QA Lead — Automation	Test strategy, harness, fixtures, CI gates, regression suite
QAIQ	QA — Image Quality	Golden diffs, blind A/B panels, skin and colour audits, artefact hunting
PERF	Performance Engineer	Benchmarks, profiling, memory ceilings, thermals, budgets

Code	Role	Owns
DEVOPS	DevOps / Release	CI/CD, signed installers, crash reporting, telemetry, auto-update
SEC	Security & Privacy	Key storage, PII/biometrics, consent, sandboxing, threat model
UX	UX / UI Designer	Flows, wireframes, review affordances, explainability, accessibility
DOC	Technical Writer	Docs, in-app help, release notes, model cards, runbooks

The two-hat rule: the agent writes code wearing one hat and reviews it wearing a different one. That single habit is what turns a code generator into an engineering organisation.

How to drive Claude Code with this bundle

1. Unzip into an empty repo and commit it as `docs/plan/`.
2. Start the session with: *"Read `CLAUDE.md` and `AGENT-TEAM.md` in full. Confirm the 9 invariants back to me. Then open `phases/PHASE-01-FOUNDATION-CATALOG-INGEST.md` and implement it task by task."*
3. For each task row say: *"Act as `SRC` (Senior Engineer — Core). Task: ... Deliverable: ... Copy the frozen contracts verbatim."*
4. Never let it start phase N+1 until phase N's Definition of Done checklist is fully ticked and its gates pass in CI.
5. Force a different hat for review: *"Now act as `QAIQ` and try to break this at 100 % zoom."*
6. When it wants to change a frozen contract, make it write an ADR first.

Non-negotiables baked into every phase

- ☐ RAW files are never modified — every edit is a recipe
- ☐ Client imagery never leaves the machine without per-project consent; your API key lives in the OS keychain, never in logs or prompts
- ☐ Every automated decision is explainable, confidence-scored and reversible

- ☐ The product works fully offline — cloud AI is an accelerator, never a dependency (cap: ~75 calls / USD 1.50 per 3,000-image wedding)
- ☐ Nothing may change what a person looks like: no body reshaping, no skin lightening, no face swapping
- ☐ Confidence bands govern autonomy: ≥ 0.98 auto · 0.90–0.98 Zero-Touch · 0.75–0.90 suggest · < 0.75 review

 Zero-Budget Build · Ship AURA for ~USD 0 and scale on revenue

 AURA — Reliability Pack · 12 Extra Blueprint Files for Error-Free Build

 AURA — Ship & Operate Pack · 8 Blueprint Files for Getting It Into Photographers' Hands

 AURA · PHASE 01 — Project Foundation, Catalog & Wedding Project Ingest · Full Build Workspace

 Engineering Constitution & 23 Agent Instruction Briefs · Enterprise-grade build law